ETFs in India: Where Are the Bugs? A Replication Study of Nifty BeES

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Abstract

This paper replicates the analytical framework of Elton et al. (2002), "Spiders: Where Are the Bugs?", to evaluate the performance and efficiency of India's first and most prominent Exchange-Traded Fund (ETF), the Nippon India ETF Nifty 50 BeES (Nifty BeES), from 2013 to 2021. We analyze its tracking error, the deviation of its market price from Net Asset Value (NAV), its performance relative to alternative investment vehicles, and the determinants of its trading volume. Our findings indicate that Nifty BeES is a highly efficient instrument. It exhibits exceptionally tight tracking of its benchmark, the Nifty 50 Total Return Index, with a minimal annual underperformance of just 0.53 basis points. Notably, unlike early-generation ETFs, efficient dividend management and low costs create a positive performance contribution that offsets minor tracking frictions. The arbitrage mechanism is effective, with price-NAV deviations showing rapid mean reversion. We quantify the "value of immediacy," finding that investors are willing to forgo 1.4 basis points in annual return for the ETF's intraday liquidity. Finally, we confirm that trading volume is driven primarily by market volatility.

Introduction: The Rise of ETFs in India

- ETFs, led by Nifty BeES, are now pivotal instruments in India's investment landscape.
- Launched in December 2001, Nifty BeES marked India's entry into passive investing.
- **Key Growth Drivers:**
  - 2015: EPFO begins equity allocations via ETFs.
  - 2018: SEBI bans upfront mutual fund commissions, leveling the playing field.
  - Government initiatives like the Bharat-22 ETF.
- **Explosive Growth:**
  - Nifty BeES AUM grew from ₹ 2 crore (2002) to ₹ 10,000 crore (2022).
  - The overall Indian ETF market saw a 45.9% CAGR over the last decade.

Introduction: Rationale for this Study

1. **Quantifying the "Value of Immediacy":**
   - ETFs offer intraday trading, unlike index funds.
   - The return differential between these vehicles quantifies the implicit cost of this intraday liquidity, a key concept in market microstructure.

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   - With over ₹ 10,000 crore in AUM and significant volume, the performance of Nifty BeES is critical for investors, regulators, and market participants.

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3. **Blueprint for Passive Investing in India:**
   - The ETF structure is now the model for future passive products. Understanding its returns and organizational form is essential.

Introduction: Choice of Nifty BeES for Replication

- We aimed to replicate the study of Elton et al. (2002) on the SPDR S&P 500 ETF (SPY).
- Among prominent Nifty 50 ETFs (ICICI Pru, UTI, etc.), Nippon India ETF Nifty 50 BeES was the best fit.
- **Criteria for Selection:**
  - Long trading history (launched in 2002).
  - High liquidity and robust secondary market activity.
  - Consistently low tracking error.
  - Transparent, in-kind creation/redemption mechanism similar to SPDRs.
- Its operational history and structural similarity make it the most appropriate choice for analyzing index replication and performance in the Indian context.

Overall Returns and Tracking Error

Headline Performance (2013-2021)

- Average Annualized Return (Market Price): **14.23%**
- Average Annualized Return (NAV): **14.23%**

This near-perfect alignment indicates that premiums/discounts have a negligible effect on long-term returns.

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Return Index (TRI)

- On average, the NAV return underperforms the benchmark index by a mere **0.53 basis points per year**.

Decomposition of the 0.53 bps Shortfall

The total underperformance is broken into two key components:

- **1. Tracking Error (–1.20 bps/year):**
  - This is the difference between the ETF's price-based return and the Nifty 50 Price Index return.
  - It reflects minor, economically insignificant frictions from rebalancing, transaction costs, etc.

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- **2. Expense and Dividend Effect (+0.66 bps/year):**
  - This is the residual component, representing the combined effect of management fees and dividend treatment.
  - **This is a crucial finding:** The positive value means efficient dividend management and low fees *more than offset* the cost drag.
  - This contrasts sharply with the "dividend drag" (a negative value) found in the original SPDR study, highlighting the structural efficiency of modern Indian ETFs.

Deviation of Price from NAV

Analysis of Pricing Efficiency

- We examine how closely the market price follows NAV by analyzing the daily proportional deviation:

$$(\text{Price}/\text{NAV}) - 1$$

- We test for persistence using the autoregressive model:

$$(\text{Price}/\text{NAV} - 1)_{t+1} = \alpha + \beta \cdot (\text{Price}/\text{NAV} - 1)_t + \varepsilon_t$$

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Table: Autoregressive Model of Price-NAV Deviations

Dependent Variable: $(Price/NAV - 1)_{t+1}$		
Variable	Coefficient	t-statistic
Intercept (α)	-0.0108	-2.20
$(\text{Price}/\text{NAV} - 1)_t$ (β)	0.6305	38.72

Interpretation of Price-NAV Deviation Results

- The slope coefficient (β) of **0.6305** is significantly less than 1, indicating a strong **mean-reverting pattern**. Deviations are corrected quickly.
- **Comparison with SPDRs:**
  - The persistence in our results ($\beta = 0.63$) is lower than that found for SPDRs ($\beta \approx 0.80$).
  - This suggests the Indian ETF market exhibits somewhat **faster correction dynamics**, with about 37% of any deviation corrected by the next day.
- The R-squared of $\sim$0.40 shows that while correction is rapid, it is not instantaneous due to market frictions.
- The negative intercept confirms the long-run expected deviation is zero, consistent with an efficient arbitrage mechanism.

Comparison with Alternatives: 1. Index Funds

Expense Ratio Comparison

- UTI Nifty 50 Index Fund: ∼17 bps p.a.
- Nippon India Index Fund – Nifty 50 Plan: ∼41 bps p.a.
- ICICI Prudential Nifty 50 Index Fund: ∼36 bps p.a.
- **Nippon BeES: only 4 bps p.a.**

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BeES has a significant cost advantage. **Other Structural Factors**

- **Transaction Costs:** Index funds use cash for subscriptions/redemptions (incurring costs), while BeES uses an in-kind mechanism (virtually no cost).
- **Securities Lending:** Index funds generate extra income (5-15 bps) by lending securities. BeES does not.

The "Value of Immediacy"

Net Performance Outcome

- Over 2013–2021, the top-performing index funds (UTI, ICICI Pru) **outperformed Nippon BeES by 1.4 basis points per year** on a pre-tax basis.

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Quantifying the Value of Intraday Liquidity

- This 1.4 bps difference can be interpreted as the price investors are willing to pay for the ETF's key benefits: intraday tradability, real-time pricing, and hedging flexibility.
- We can state that sophisticated investors value immediacy at **at least 1.4 basis points per year**.
- This makes the value of immediacy in the Indian market remarkably affordable.

Comparison with Alternatives: 2. Futures

The Synthetic Replication Strategy

- An alternative is to hold short-term T-bills plus Nifty 50 futures contracts. We compared the returns of this strategy to BeES and the benchmark.

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Annualized Return Comparison (2013-2021)

- Nifty 50 TRI (Benchmark): **14.7150%**
- Nippon BeES: **14.1762%**
- Synthetic Futures + T-bill Strategy: **12.6787%**

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The futures strategy **massively underperformed** by ∼150-204 bps. This is likely due to futures consistently trading at a premium to fair value. Despite theoretical appeal, Nifty BeES is the superior and simpler passive vehicle.

Determinants of Volume: Price-NAV Deviation

Does Arbitrage Drive Volume?

- We test if the price deviation on day t predicts trading volume on day $t+1$.

$$\text{Volume}_{t+1} = \alpha + \beta \cdot (\text{NAV}_t - \text{Price}_t) + \varepsilon_t$$

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- Slope (β) = -3.75 million (t-stat = -41.87)

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Interpretation: There is a strong, statistically significant **negative relationship**.

- When BeES is at a **premium** (Price ¿ NAV), next-day volume **declines**.
- When BeES is at a **discount** (Price ¡ NAV), next-day volume **rises**.

Determinants of Volume: Market Volatility

Does Market Uncertainty Drive Volume?

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Interpretation: The relationship is large, positive, and highly significant. BeES volume rises sharply during periods of high market volatility, likely due to increased trading for hedging and portfolio rebalancing by all types of investors.

Determinants of Volume: The Full Picture

Multivariate Model

- We run a multiple regression including both price deviation and volatility as joint predictors of next-day volume.

Table: Multivariate Regression of ETF Trading Volume

Dependent Variable: Volume$_{t+1}$		
Variable	Coefficient	t-statistic
Intercept	30,743	0.58
$(\text{NAV} - \text{Price})_t$	8,195	0.05
Volatility_t	64.47 million	17.47
R-squared	**0.110**	

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Key Finding: Volatility alone is the dominant driver of volume.

Conclusion: Summary of Key Findings I

This replication study finds that the Nippon India ETF Nifty 50 BeES is a remarkably robust and efficient investment vehicle, free from the structural flaws or "bugs" that were a concern in the early days of ETFs.

1. **High Tracking Fidelity:**
   - Nifty BeES tracks its benchmark with near-perfect accuracy.
   - Crucially, modern structural advantages like efficient dividend handling and low costs have eliminated the "dividend drag" seen in older ETFs, creating a net positive contribution to performance.

2. **Efficient Arbitrage:**
   - The ETF's market price stays closely tethered to its NAV.
   - The arbitrage mechanism is effective, ensuring that deviations are small and corrected more rapidly than in the market studied by Elton et al. (2002).

3. **Quantified Value of Immediacy:**

Conclusion: Summary of Key Findings II

- The ETF provides intraday liquidity and real-time pricing at a very low implied cost of just 1.4 basis points per year when compared to top-performing index funds.

4. **A Key Tool for Risk Management:**
   - Trading activity in Nifty BeES is overwhelmingly driven by market volatility, cementing its role as a primary instrument for investors to navigate and manage market-wide risk.

Final Verdict

In the Indian context, Nifty BeES has proven to be a highly successful financial innovation, reinforcing its role as a responsive, efficient, and structurally sound cornerstone of the passive investment landscape.